CLASS 9 SCIENCE · QUICK NOTES

CHAPTER 7

Work, Energy, and Simple Machines

Work · Work–energy theorem · Kinetic & potential energy · Conservation · Power · Pulley, inclined plane, lever

THE CHAPTER IN 30 SECONDS

Work is force acting through a displacement (W = F × s). **Energy** is the capacity to do work, and work done on an object equals the **change in its energy**. **Mechanical energy = KE + PE** stays constant when no other external force acts, a shortcut that solves many problems. **Power** is how fast work is done. **Simple machines** (pulley, inclined plane, lever) change the force needed but **never reduce the total work**.

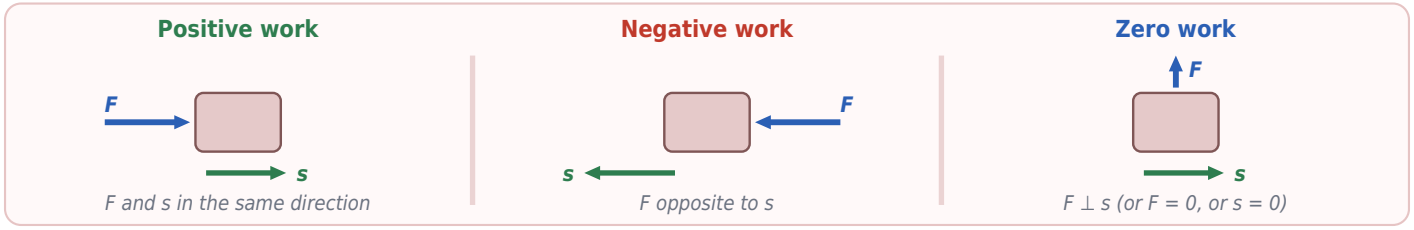
1 Work done by a constant force

NCERT 7.1

W = F × s
s = displacement in the direction of the force

1 J = 1 N × 1 m
= 1 kg m² s⁻² · SI unit of work: joule (J)

- Always name the **force** and the **object** it acts on. Work has **no direction**, only a + or − sign.
- More force over the same distance, or the same force over more distance → more work (3 bags together at 1 m = 3 W; 1 bag to 3 m = 3 W).
- **Force-displacement graph:** work = **area under the graph** (10 N × 1 m = 10 J). This works even when the force is not constant.



Work is	When	Examples
Positive	displacement is in the direction of the force	pushing a wheelchair; lifting a dumbbell up
Negative	displacement is opposite to the force	goalkeeper stopping a ball; lowering a dumbbell; friction on a sliding object
Zero	F = 0, or s = 0, or force ⊥ displacement	pushing a rigid wall; weightlifter holding a barbell steady; carrying a box while walking (upward force ⊥ motion)

✎ **Goalkeeper (Example 7.2)**
F = 200 N, hand moves back 0.15 m opposite to the force → W = 200 × (−0.15) = **−30 J**.

★ **MEMORY HOOK**
"Tired ≠ work." Pushing a wall makes you tired (muscles use internal energy) but the wall does not move, so work done = 0.

2 Work–energy theorem and forms of energy

NCERT 7.2, 7.3

work done on an object = change in its energy
positive work → energy increases · negative work → energy decreases · unit: joule (J)

- **Energy** = capacity to do work. The theorem also holds for systems and for non-constant forces.
- **Carrom shot:** striker does positive work on the white coin (coin gains energy); the coin does negative work on the striker (striker loses energy).
- Energy also moves by **heat**, **radiation**, sound waves and electric circuits, not only by mechanical work. SI unit named after James Prescott Joule.

Form	Meaning	Form	Meaning
Mechanical	due to motion or position	Electrical	related to charges (position or motion)
Thermal	makes things warm or hot	Nuclear	stored in the nuclei of atoms
Light	allows us to see	Chemical	stored in fuels and food (chemical bonds)
Sound	vibrations of air or other molecules		

Conversions: bulb: electrical → light · water heater: electrical → thermal · food: chemical → mechanical · ringing bell: mechanical → sound.

3 Kinetic energy

NCERT 7.4.1

$K = \frac{1}{2}mv^2$

energy of motion · unit: joule

WHERE IT COMES FROM

$v^2 = u^2 + 2as \rightarrow s = (v^2 - u^2) \div 2a$. Then $W = F \cdot s = ma \cdot s = \mathbf{\frac{1}{2}m(v^2 - u^2)}$. For $u = 0$ the work done is the final kinetic energy: $K = \frac{1}{2}mv^2$.

- No direction; zero for an object at rest. Positive work → KE rises, negative work → KE falls, zero work → KE unchanged.
- **Speed × 2 → KE × 4** (KE ∝ v^2). Speed × 3 → KE × 9.

✎ **Cricket ball (7.5)**

0.2 kg at 154.8 km h^{-1} = 43 m s^{-1} → $K = \frac{1}{2} \times 0.2 \times 43^2$ = **184.9 J**.

✎ **Aircraft carrier (7.6)**

15000 kg jet, stopping force 367500 N over 100 m. Work by wire = $-367500 \times 100 = -\frac{1}{2} \times 15000 \times v^2 \rightarrow v^2 = 4900 \rightarrow$ **v = 70 m s^{-1} = 252 km h^{-1}**.

4 Potential energy

NCERT 7.4.2

$U = mgh$

gravitational PE near the Earth's surface · unit: joule

WHERE IT COMES FROM

To raise an object slowly through h, apply F = mg, so work W = mg × h. That work is stored as potential energy: U = mgh.

- **Potential energy** = energy stored by **deformation** (stretched rubber band, bent bow, slingshot, compressed spring) or by the **relative positions** of objects in a system (magnets, electric charges, Earth and ball).
- Stored energy is released as kinetic energy (a released bow shoots the arrow). Greater height → greater PE (a ball dropped from 2 m makes a deeper dent in sand than from 1 m).
- g decreases far from the Earth, so U = mgh is only valid near the surface. Not every internal force stores energy: work against **friction** is not stored.

✎ **Fielder's ball (Example 7.7)**

0.2 kg ball, 10 m above the ground, g = 10 m s^{-2} → U = mgh = 0.2 × 10 × 10 = **20 J**.

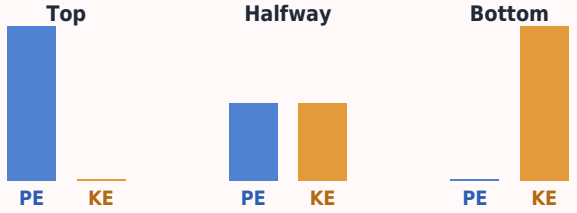
5 Conservation of mechanical energy

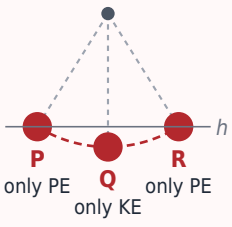
NCERT 7.4.3

mechanical energy = KE + PE = constant

when no other external force acts (e.g. a freely falling object)

Free fall: PE falls, KE rises, total stays mgh





Falling object	Potential energy	Kinetic energy	Mechanical energy
Point A (top, at rest)	mgh	0	mgh
Point B (after time t): v = gt	$mgh - \frac{1}{2}mg^2t^2$	$\frac{1}{2}mg^2t^2$	mgh (same)

- **Slide:** PE at the top becomes KE at the bottom: ½mv² = mgh → **v = √(2gh)**. Speed depends only on the height h, not on the mass or the shape of the slide.
- **Escape ramp (Example 7.9):** truck 10000 kg at 20 m s⁻¹ has KE = 2,000,000 J. On a 30° ramp it rises d ÷ 2 (PE = 50000 d) and sand does −50000 d: 2,000,000 = 50000 d + 50000 d → **d = 20 m**.

★ MEMORY HOOK

Energy bookkeeping: what PE loses, KE gains; the total stays the same. Real motion loses some energy to heat (friction, air), so a pendulum finally stops.

6 Power

NCERT 7.5

P = W ÷ t
rate of doing work

1 W = 1 J s⁻¹
SI unit: watt (W) · 1 horsepower (hp) = 746 W

The same work done in **less time needs more power** (running upstairs vs walking up). Unit named after James Watt (steam engine).

✎ **Weightlifter (7.10)**
75 kg lifted 2 m in 5 s. W = mgh = 75 × 10 × 2 = 1500 J → P = 1500 ÷ 5 = **300 W**.

✎ **Car (7.11)**
1000 kg, 0 → 72 km h⁻¹ (20 m s⁻¹) in 10 s. W = ½ × 1000 × 20² = 200000 J → P = 200000 ÷ 10 = **20000 W**.

7 Simple machines

NCERT 7.6

MA = load ÷ effort
effort = force we apply · load = force to overcome

BIG IDEA

Simple machines make a task easier by changing the **size or direction** of the force. They do **not** reduce total work and do **not** create energy (work in = useful work out, ignoring friction).

Machine	How it helps	Mechanical advantage
Fixed pulley	only changes the direction of effort (pull down instead of lifting up)	= 1
Movable pulley / pulley system	reduces the effort needed (cranes, lifts)	> 1
Inclined plane	smaller force over a longer distance	L ÷ h
Lever	small effort acting through a long effort arm	effort arm ÷ load arm

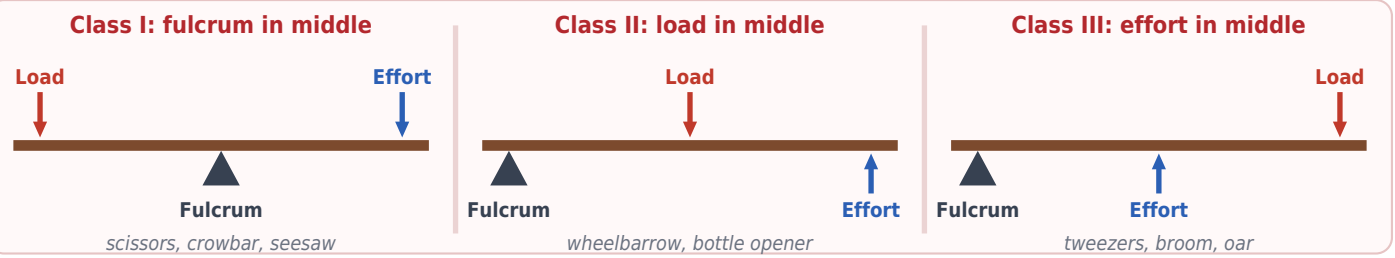
Inclined plane. Moving the load at constant speed: F′ × L = mgh, so **MA = mg ÷ F′ = L ÷ h**. A longer (gentler) ramp needs less force but a longer push; the work is the same. Why: winding hill roads, ladders. *Example 7.12:* ramp 30 cm high with 40 cm base → L = 50 cm → MA = 50 ÷ 30 = **1.67**.

Inclined plane: MA = L ÷ h



LEVER

A **rigid bar** turning about a **fulcrum**. Balance: **effort × effort arm = load × load arm** ($F_1d_1 = F_2d_2$: work in = work out). **MA = effort arm ÷ load arm**. A small effort moves a large distance; the load moves a small distance. *Seesaw (7.13):* 15 kg at 2 m balances 30 kg at L: 15 × 2 = 30 × L → **L = 1 m** (seat D).



★ MEMORY HOOK

F-L-E → 1-2-3: the item in the **middle** names the lever class. **Gain in force = loss in distance**: no machine reduces total work.

★ Quick revision

one-page summary

Quantity	Formula	Unit
Work	W = F × s (s along the force)	joule (J) = N m
Work–energy theorem	work done = change in energy	J
Kinetic energy	K = ½mv²	J
Potential energy (gravity)	U = mgh	J
Conservation	KE + PE = constant (free fall: v = √(2gh) from height h)	J
Power	P = W ÷ t	watt (W) = J s^{-1} ; 1 hp = 746 W
Mechanical advantage	MA = load ÷ effort ; ramp L ÷ h ; lever effort arm ÷ load arm	no unit
Lever balance	effort × effort arm = load × load arm	

10 POINTS TO REMEMBER

1. Work is done only when the force causes displacement **in its direction**; unit joule.
2. Work can be **positive, negative or zero** (F = 0, s = 0, or force perpendicular to displacement).
3. Work done on an object = **change in its energy**.
4. Kinetic energy ½mv² depends on **v²**: double the speed → 4 × the KE.
5. Potential energy = energy stored by deformation or relative position; gravitational U = mgh.
6. Without other external forces, **KE + PE is constant**; a falling object turns PE into KE.
7. Speed at the bottom of a frictionless slide = √(2gh), independent of mass and shape.
8. Power = work ÷ time; same work in less time = more power.
9. MA = load ÷ effort. Fixed pulley MA = 1; ramp MA = L ÷ h; lever MA = effort arm ÷ load arm.
10. Machines change force or direction but **never reduce total work**.

✗ COMMON MISTAKE

- Saying work is done when you push a wall or hold a weight still.
- Forgetting negative work (friction, a goalkeeper stopping a ball).
- Using kg, m and J mixed with g, cm: convert to SI before substituting. Remember KE depends on v², not v.
- Thinking a machine gives “free” work, or turning the lever MA upside down (it is effort arm ÷ load arm).

TEST YOURSELF (cover the green column)

Q1. A 5 kg bag is lifted slowly 2 m (g = 10 m s^{-2}). Work done on the bag?	**W = mgh = 5 × 10 × 2 = 100 J**
Q2. What happens to the kinetic energy if the speed triples?	**It becomes 9 times**
Q3. A girl carries a box horizontally at constant speed. Work done by the upward force?	**Zero (force ⊥ displacement)**
Q4. A machine does 600 J of work in 20 s. Its power?	**P = 600 ÷ 20 = 30 W**
Q5. On a seesaw, 15 kg sits 2 m from the fulcrum. Where must 30 kg sit to balance?	**1 m from the fulcrum (opposite side)**